Separating Risk Signal from Volume in EU Food-Safety Alerts: A Two-Layer Network Analysis of RASFF Notifications with Null-Model and Out-of-Sample Validation

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Technical report, October 2026. Code, data, and results: github.com/GeneTheStoic/rasff-network

Abstract. Food-safety alert systems are used to set monitoring priorities, but raw notification counts mix product-specific risk with trade volume, inspection intensity, and sampling noise. I build a two-layer network (origin country to product category, product category to hazard class) from 13,068 food notifications in the EU Rapid Alert System for Food and Feed (RASFF) between January 2023 and October 2026, and ask three questions: which nodes are central, how stable their rankings are, and which links are stronger than volume alone predicts. Centrality rankings are stable across years (Spearman ρ = 0.90–0.94) and under 1,000 bootstrap resamples at the top, but not near the top-10 cut-off. Testing every link against a fixed-margin null model with false discovery rate control, and validating out of sample, separates the two readings of the data: 86% of origin–product links validated on 2023–2025 remain over-represented in 2026, against 30% of the other links. A test of coupling between layers shows that the largest node, Türkiye's fruit and vegetable notifications, is driven by mycotoxins (2.6 times expected) and not by pesticide residues (0.72 times expected), a result that holds in 2026.

1. Motivation

Risk managers read alert data as evidence of where to look next. Two problems make that reading fragile. First, counts and centrality measure how often a node appears, which depends on how much is traded and how much is inspected, not only on how risky it is. Second, rankings built from a few thousand notifications carry sampling uncertainty that is rarely reported. A network view is useful here because the same product connects the countries that ship it to the hazards found in it, so the question of where risk sits becomes a question about the structure of links and not only about the size of nodes.

This report addresses three questions. **Q1:** which origin countries, product categories, and hazard classes are central in EU notifications? **Q2:** how stable are those rankings under resampling and over time? **Q3:** which links, and which combinations across layers, are stronger than the volume of their endpoints predicts, and do they persist out of sample?

2. Data

The data are the public RASFF Window export of 22 October 2026: 19,801 notifications from 2 January 2023 to 21 October 2026. I keep the 13,068 food notifications with at least one coded hazard (2,998 in 2023; 3,519 in 2024; 3,549 in 2025; 3,002 in 2026 up to 21 October), excluding feed, food-contact materials, and cases without a coded hazard. The hazard field lists one or more hazards per notification in the form *name - {class}*; it is parsed into separate records, and notifications with several origin countries are split by country. The sample covers 141 origin countries, 33 product categories, 28 hazard classes, and 931 distinct hazards; 57.3% of notifications carry a *serious* risk decision.

3. Methods

3.1 Network construction

Layer 1 links origin countries to product categories and layer 2 links product categories to hazard classes. Edge weights count notifications, with each notification counted once per origin in layer 1 and once per hazard class in layer 2. Product categories are shared by both layers, which connects them into a 202-node network with 1,443 weighted edges.

3.2 Centrality and communities

I compute weighted degree (strength) and PageRank on the combined network (Brin and Page, 1998). Communities of products are found with the Louvain method (Blondel et al., 2008) on the weighted projection of layer 2, in which two products are linked when they share hazards.

3.3 Uncertainty and temporal stability

The bootstrap resamples notifications with replacement (1,000 replications), so that every origin and hazard record of a notification moves together, rebuilds the network, and recomputes PageRank (Efron and Tibshirani, 1993). For each node I report the median, a 90% percentile interval, and the share of replications in which it ranks in the top 10 of its type. Temporal stability is the Spearman correlation of PageRank between consecutive years.

3.4 Null models, false discovery rate, and out-of-sample validation

Links within a layer. For a link between a and b, let N be the number of notification records in the layer and $n(a)$ and $n(b)$ the margins. If the two margins are independent, the link count follows a hypergeometric distribution with expected value $n(a)\ n(b)\ /\ N$, and the p-value is the probability of a count at least as large as the one observed. This is the statistically validated network approach of Tumminello et al. (2011).

Coupling between layers. For a triple of origin o, product p, and hazard h, the null hypothesis is that origin and hazard are independent given the product: within product p, the count follows a hypergeometric distribution with population $n(p)$ and margins $n(o, p)$ and $n(p, h)$, so the expected count is $n(o, p)\ n(p, h)\ /\ n(p)$. Under this null the two layers interact only through the shared product node; a rejection is an origin-specific hazard profile that neither layer shows on its own.

Multiple testing and validation. Within each family of tests, links are validated at a Benjamini–Hochberg false discovery rate of 1% (Benjamini and Hochberg, 1995). Validation uses 2023–2025 only (10,066 notifications). The 2026 notifications (3,002) are held out: for every link I recompute the expected count from 2026 margins and report the share of links observed above expectation and the pooled ratio of observed to expected counts.

4. Results

4.1 Structure of the network (Q1)

Three hazard classes dominate: pesticide residues (PageRank 0.051), pathogenic micro-organisms (0.049), and mycotoxins (0.034); the next class, heavy metals, has 0.011. Among products, fruits and vegetables (0.113), nuts, nut products and seeds (0.063), and poultry meat (0.043) lead; among origins, Türkiye (0.023) is followed by Poland and India (0.014 each). The heaviest edges are fruits and vegetables to pesticide residues (2,168 notifications), poultry meat to pathogenic micro-organisms (1,414), and nuts to mycotoxins (1,204); on the origin side, Türkiye to fruits and vegetables (1,110) and Poland to poultry meat (661). Products split into two communities by the hazards they share: 15 categories, mostly of animal origin (poultry and other meat, eggs, milk, molluscs, crustaceans), linked to microbiological hazards, and 18 plant-based and processed categories (fruits and vegetables, nuts, spices, cereals), together with fish, linked to chemical hazards.

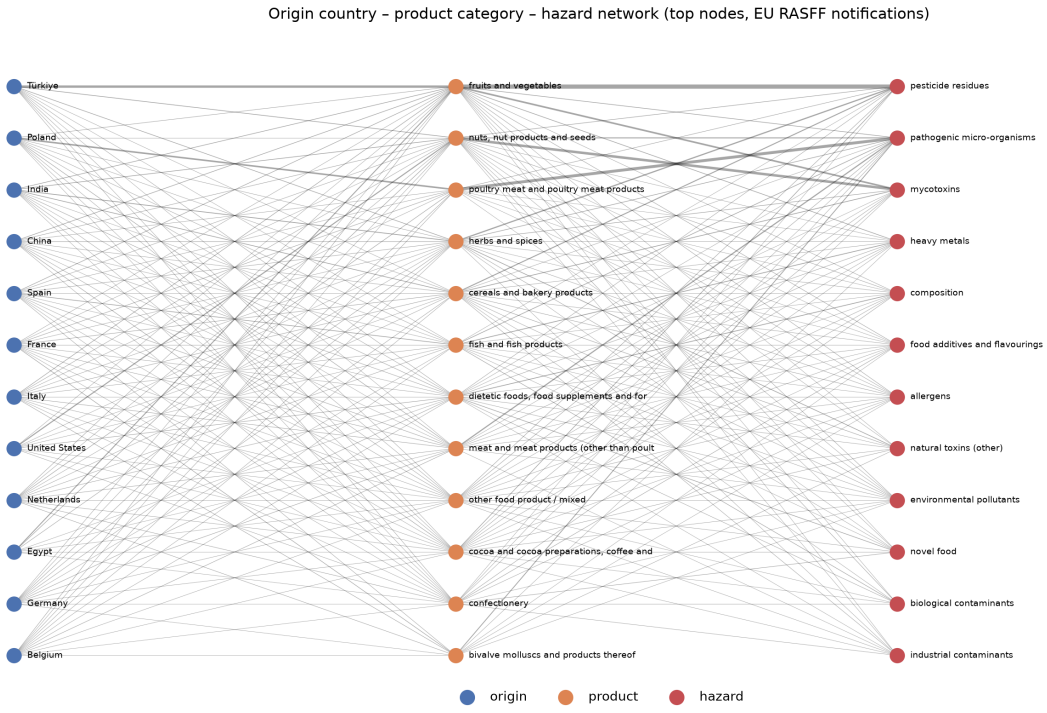


Figure 1. Two-layer network, top 12 nodes of each type ordered by PageRank. Edge width is proportional to the number of notifications.

4.2 Stability of the rankings (Q2)

The top of every ranking is robust. The nine leading hazard classes and the eight leading product categories rank in the top 10 in all 1,000 bootstrap replications, and rankings are stable from year to year (Spearman ρ = 0.922 for 2023–2024, 0.902 for 2024–2025, and 0.936 for 2025–2026). The boundary is not: *novel food* stays in the hazard top 10 in 90.1% of replications while *biological contaminants* enters in 9.5%, and *cocoa, coffee and tea* stays in the product top 10 in 94.4% while *confectionery* enters in 7.3%. A priority list built from a single ranking should therefore treat its last positions as uncertain.

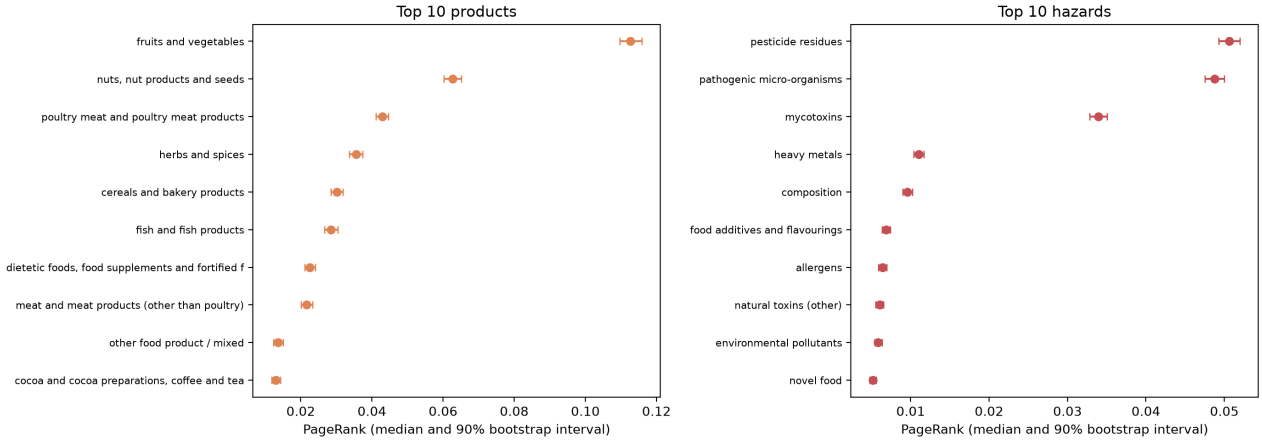


Figure 2. Top 10 product categories and hazard classes: median PageRank and 90% bootstrap interval (1,000 replications).

4.3 Links stronger than volume predicts (Q3)

Table 1 reports the null-model results. Validated links are a minority in every family, and they are the ones that persist: in 2026, 86% of validated origin–product links and 82% of validated product–hazard links are again above expectation, with pooled ratios of 3.51 and 2.92, while the links that were not validated fall below expectation (0.66 and 0.48).

Table 1. Null-model validation (hypergeometric tests, Benjamini–Hochberg FDR 1%) and out-of-sample check on 2026. O/E is the pooled ratio of observed to expected counts, with expectations recomputed from 2026 margins.

Test family	Links	Validated (2023–2025)	Above expectation in 2026: validated	Above expectation in 2026: not validated
Origin–product	970	135	86% (O/E 3.51)	30% (O/E 0.66)
Product–hazard	350	77	82% (O/E 2.92)	25% (O/E 0.48)
Origin–product–hazard	2,341	144	75% (O/E 1.70)	32% (O/E 0.94)

Several of the largest raw links are explained by volume: Türkiye to nuts (146 notifications, 0.82 of expected), India to fruits and vegetables (111, 0.61), and fruits and vegetables to mycotoxins (498, 1.11) are not over-represented once the margins are taken into account. Smaller links carry specific signals: Ecuador to crustaceans (82 notifications, 33 times expected), Pakistan to cereals (194, 11 times), fish to parasitic infestation (43, 19 times), and cephalopods to heavy metals (24, 18 times).

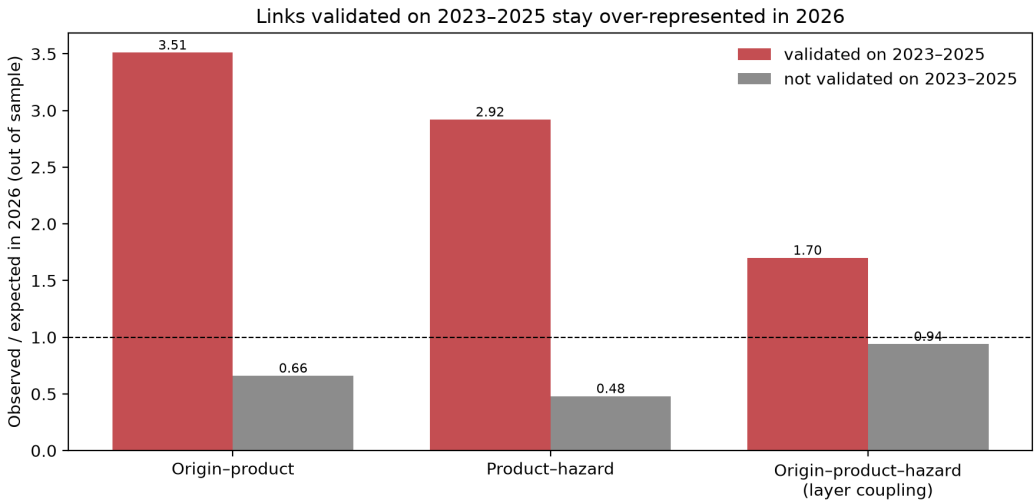


Figure 3. Out-of-sample check: pooled ratio of observed to expected counts in 2026 for links validated and not validated on 2023–2025. The dashed line marks the null expectation.

4.4 Coupling between layers

The coupling test changes the reading of the largest node in the network. Türkiye's fruit and vegetable notifications are the largest origin–product link and fruits and vegetables are most often notified for pesticide residues, so a reading by layer suggests pesticides. Conditional on the product, however, the triple Türkiye, fruits and vegetables, pesticide residues is below expectation (387 notifications against 541 expected, ratio 0.72), while the triple with mycotoxins is 2.57 times expected (416 against 162). Both results hold in the held-out year: 182 mycotoxin notifications against 70 expected in 2026, and 84 pesticide notifications against 162 expected. Other validated triples include China, poultry meat, pesticide residues (26 notifications, 17 times expected), Vietnam, fish, residues of veterinary medicinal products (20, 12 times), Peru, fruits and vegetables, heavy metals (22, 9 times), and Indonesia, herbs and spices, mycotoxins (27, 8 times). Across all triples, validated combinations are above expectation in 2026 in 75% of cases, against 32% for the rest.

5. Limitations

RASFF notifications record what is inspected and reported, not the prevalence of contamination; central nodes partly reflect inspection intensity, including reinforced border controls on specific products and origins. The null models condition on observed margins, so they separate links from the volume of their endpoints but do not correct for inspection effort itself, and the hypergeometric tests treat links as separate tests with fixed margins, without modeling overdispersion. Notifications with several origins contribute one record per origin. Hazards are analyzed at the class level, which is coarse; the 931 hazard names are available in the cleaned data but are sparse. Counts do not include consumption or exposure, so this is a screening and prioritization exercise, not a risk assessment, and 2026 is a partial year.

6. Extensions

Four extensions follow directly. First, replace link-by-link tests with models estimated jointly across layers, such as weighted stochastic block models or latent space models for multi-layer networks, so that the dependence between layers is modeled rather than tested one triple at a time. Second, normalize by inspection effort and trade volume to separate surveillance intensity from risk. Third, add an exposure layer linking product categories to consumption data, for example national food consumption surveys, so that priorities are weighted by how much of each food a population eats. Fourth, carry the uncertainty of every network quantity through to the final priority list, validated over time, so that risk managers can see which priorities are stable.

References

Benjamini, Y., and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, P10008.

Brin, S., and Page, L. (1998). The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems*, 30(1–7), 107–117.

Efron, B., and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman and Hall.

European Commission. RASFF Window, Rapid Alert System for Food and Feed. https://webgate.ec.europa.eu/rasff-window/screen/search (accessed 22 October 2026).

Tumminello, M., Micciché, S., Lillo, F., Piilo, J., and Mantegna, R. N. (2011). Statistically validated networks in bipartite complex systems. *PLoS ONE*, 6(3), e17994.

Code and reproducibility. All steps (cleaning, network construction, centrality, bootstrap, null models, and figures) are in five Python scripts at github.com/GeneTheStoic/rasff-network and reproduce every number in this report from the public RASFF export.